

Generation Interconnection System Impact Study Report

For Queue Project Q19-1360

Floraday 138 kV
February 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1360. The Interconnection Customer proposes a 136.3 MW solar photovoltaic facility interconnecting at the Floraday 138 kV substation in the Virginia area. The request was received on December 15, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 482 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$326,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Deen Pt to Goldfield 138.0 kV: 108.3% loading (214 MVA) following the outage of the Mccartey to N Tower Tap 345.0 kV; DFAX = 41.1%.

For the loss of Mitchell Dam to Haggin 345.0 kV, the Turner Rd to Tyler 138.0 kV Line loads to 107.3% of its 271 MVA rating. Project DFAX: 8.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Deen Pt to Goldfield Substation 138.0 kV	Reconductor line	\$1,225,000	15% (\$182,000)
Turner Rd to Tyler 138.0 kV Line	Reconductor line	\$1,868,000	8% (\$144,000)
	Total allocated to project		\$326,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.